

Principal Component Analysis (PCA) is a **dimensionality reduction** technique that transforms a dataset with potentially correlated variables into a set of linearly uncorrelated variables called **principal components (PC)**.

- PCA (Principal Component Analysis) itself does not reduce dimensionality — it **transforms** the data into a new coordinate system.
- Dimensionality reduction happens only when we choose to keep only the top **k** principal components. (cumulative variance explained > 50% or PCs with eigenvalues >1)

Algorithm:

1. Standardize the data
 - To ensure that each feature contributes equally because PCA is sensitive to scale of variables.
 - Or variables with larger variances (scales) will dominate the principal components, leading to misleading results.
2. Compute the covariance matrix
 - To measure how features vary together.
 - If X1 and X2 are highly correlated, their covariance will be large, indicating that PCA may combine them into one component.

$$C = \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix}$$

3. Calculate eigenvalues and eigenvectors of the covariance matrix
 - Eigenvectors define the new directions in the transformed space.
 - Eigenvalues tell you how much variance is captured in each direction.
 - This is the core mathematical transformation in PCA.
 - $\det(C - \lambda I) = (1 - \lambda)^2 - c^2$ (To find the eigenvalues of matrix C)

- e.g. $C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $\det(C - \lambda I) = (1 - \lambda)^2 - 4 = 0$, $\lambda = -1$ or 3 .

4. Sort eigenvectors by decreasing eigenvalues
 - To rank the principal components by importance.
5. Transform the original data
 - To project the data onto the new PCs.